Joint Trajectory and Passive Beamforming Optimization in IRS-UAV Enhanced Anti-Jamming Communication Networks

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Abstract: This paper investigates the anti-jamming communication scenario where an intelligent reflecting surface (IRS) is mounted on the unmanned aerial vehicle (UAV) to resist the malicious jamming attacks. Different from existing works, we consider the dynamic deployment of IRS-UAV in the environment of the mobile user and unknown jammer. Therefore, a joint trajectory and passive beamforming optimization approach is proposed in the IRS-UAV enhanced networks. In detail, the optimization problem is firstly formulated into a Markov decision process (MDP). Then, a dueling double deep Q networks multi-step learning algorithm is proposed to tackle the complex and coupling decision-making problem. Finally, simulation results show that the proposed scheme can significantly improve the anti-jamming communication performance of the mobile user.

Keywords: intelligent reflecting surface; unmanned aerial vehicle; deep reinforcement learning; trajectory optimization

I. INTRODUCTION

The application of intelligent reflecting surface (IRS) in anti-jamming is an infant but promising technology [1]. By introducing IRS into networks comprising active components only, the performance of the new IRS-aided hybrid wireless networks can be further improved [2–7]. On the other hand, unmanned aerial vehicle (UAV) has the characteristics of rapid deployment, flexible movement and low cost. The application of UAVs in wireless communication has been extensively studied [8–11]. Thus, the potential advantages of UAV and IRS motivate us to consider combining them together. In this paper, the IRS is mounted on the UAV (IRS-UAV) to assist anti-jamming communication with trajectory and passive beamforming joint optimization.

IRS is composed of many reflection unit cells, which can control the phase and amplitude of the reflection signal [12]. Through phase adjustment, combining the reflected signal and direct signal at the receiver can achieve the signal superimposition/elimination [13]. Also, IRS can operate in the full-duplex mode without self-interference [14]. When we mount IRS on UAV, the ground-to-air channel suffers from much lower channel attenuation than the ground channel, which can significantly reduce the energy loss of passive reflection. Additionally, the mobility of UAVs can give IRS the ability to adjust the deployment position, which can fully exploit the performance in a dynamic environment.

However, the optimization problem of IRS-UAV in the anti-jamming scenario also faces many challenges. First, when we utilize IRS in the anti-jamming scenario, the optimization of passive beamforming is far more complex, i.e., the beamforming of IRS needs

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to strike a balance between enhancing the useful signals and mitigating the jamming signals. Secondly, the jammer information is unknown to the mobile communication transceiver in this scenario. This unknown and dynamic environment makes optimization challenging to be solved by conventional optimization approaches. Thirdly, the combination of IRS and UAV makes the trajectory and passive beamforming severely affect each other. Thus the variables of joint optimization are highly coupled.

In order to tackle the challenges of joint optimization, the anti-jamming communication problem is formulated into the Markov decision process (MDP) by adequately designing the state, action and reward. Then we propose the dueling double deep Q networks (3-DQN) multi-step learning algorithm, which provides IRS-UAV the ability to learn the best planning for trajectory and passive beamforming in a dynamic unknown environment.

The main contributions of this paper are summarized as follows.

- Considering the dynamic deployment of IRS-UAV, the anti-jamming problem is formulated into an MDP. We significantly reduce the action space by utilizing the reflection channel mode of IRS, especially when the large number of IRS is.
- In the light of unknown jammer and the mobile user, the problem is solved by a dueling double deep Q networks multi-step learning algorithm. The proposed algorithm has an excellent performance by improving the traditional deep Q networks with dueling layers, double networks, and multi-step.
- Simulation results are presented and analyzed to verify the effectiveness of the proposed scheme. Results demonstrate that this IRS-UAV-assisted anti-jamming scheme can provide outstanding anti-jamming communication performance with trajectory and passive beamforming joint optimization.

The rest of this paper is organized as follows. Section VI summarizes some related work. Section V gives the system model and the problem formulation. Section IV proposes the dueling double deep Q networks multi-step learning algorithm. Section III presents the simulation results and analyses. Section II is the conclusion.

Notions: $\mathbb{C}^{x \times y}$ denotes the space of $x \times y$ complex-valued matrices. The superscript $(\cdot)^H$ denotes the transpose and Hermitian conjugate of the matrix. $\mathbb{E}(\cdot)$ denotes the statistical expectation. $diag(\cdot)$ denotes the diagonal matrix.

II. RELATED WORK

With the rapid development of communication technology, the jamming attack has become an enormous threat to wireless communication [15–17]. The existing anti-jamming works mainly focused on the power domain, frequency domain and space domain [18]. However, the above approaches are challenging to deal with high-power jamming, mainly when the jammer can track the changes of the communication frequency band. For example, the anti-jamming approaches of the power domain can not deal with jamming with strong power [19–21], and the approach of frequency hopping in the frequency domain can not avoid the influence of tracking jamming when the capabilities of both parties are equal. [22–25]. Beamforming in the space domain can use multi-antenna gain to significant anti-jamming effects and increases the complexity of communication systems and energy consumption. Thus this paper proposes a promising IRS-UAV-assisted anti-jamming scheme to deal with the suppression tracking jamming.

The problem of IRS-assisted anti-jamming in the static scene has been studied [26]. H. Yang et al. focused on the optimization problem of power allocation at base station and beamforming at IRS to satisfy legitimate users' quality of service requirements[27]. However, when the user is mobile, the fixed deployment of IRS can not fully exploit the anti-jamming performance. Hence we consider the dynamic deployment of IRS instead of fixed location IRS, which leads to the dynamic optimization problem.

In this paper, we mainly consider the joint optimization of trajectory and beamforming. In [28–33], such a problem has been investigated. There are two main differences in our work. First, we consider the trajectory and beamforming of IRS-UAV in a dynamic environment. The related works focus on the trajectory of the UAV while the IRS is fixed in a specific location. The coupling relationship of IRS and UAV in this paper increases the complexity of the joint optimization problem. Secondly, we study the problem in the

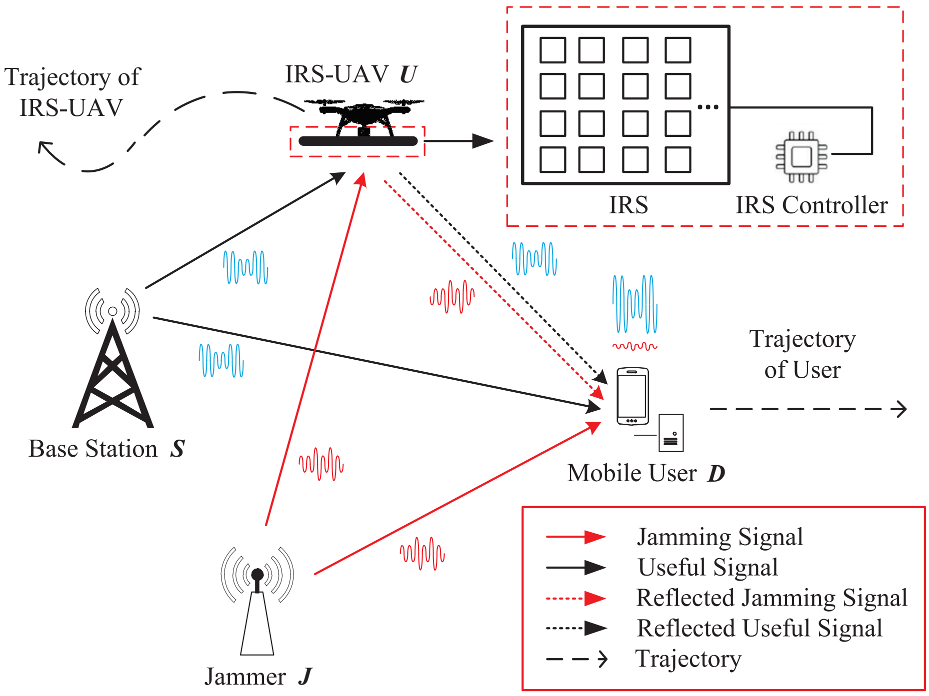

Figure 1. *The structure of IRS-UAV anti-jamming scheme.*

jamming scenario with imperfect information of jammer. The optimization needs to enhance the strength of the communication signals and reduce the influence of jamming signals.

III. SYSTEM MODEL AND PROBLEM FORMULATION

As shown in Figure 1, we consider a general single-input single-output wireless communication system in this paper. There are legal communication nodes: base station S and mobile user D, and malicious jammer node J sending the omnidirectional signal to interfere with communication nodes on the ground. The IRS-UAV node U flies in the air to assist in anti-jamming communication. Assuming the jammer is an intelligent jammer, which can hide the location and power itself. Besides, the user is moving within specified scope, and its trajectory is determined by task requirements that could not be predicted. Therefore, the IRS-UAV-assisted anti-jamming communication system is in a dynamic unknown environment.

The joint optimization objective of trajectory and beamforming is to maximize the achievable communication rate at the mobile user. The total duration T is divided into N time slots $T = N\tau$ [34], and the set of time slots is defined as $\mathcal{N} = \{1, ..., N\}$. When N is big enough, the length of time slot τ is especially small, which means that the position of UAV and the phase shift of IRS are static in time slot n. Supposing the scene is an open area, the direct link between S and D and the direct link between J and D exist. Notably, we assume all the channels follow the quasi-static flat fading channel. This assumption is ideal since it is less capable of handling channel estimation and frequency-selective fading, which is not the focus of our research [35]. The IRS can adjust the phase and amplitude of each beam reflected by the reflecting unit cells adapting to the dynamic channel. By focusing on the adjusted beam, the anti-jamming performance of the mobile user can be significantly improved.

3.1 Dynamic Movement Model

In the dynamic scene, IRS-UAV is assumed to fly at a fixed height H, and the height of ground nodes is 0. Thus, only the movement of the two-dimensional plane is considered. Denote the coordinates of S, J, D, and U in slot n as $\mathbf{W}_S = \{X_S, Y_S\}$, $\mathbf{W}_J = \{X_J, Y_J\}$, $\mathbf{W}_D[n] = \{X_D[n], Y_S[n]\}$ and $\mathbf{W}_U[n] = \{X_U[n], Y_U[n]\}$. The distance between S and D in slot n is expressed as

$$d_{SD}[n] = \sqrt{(X_S - X_D[n])^2 + (Y_S - Y_D[n])^2}. \quad (1)$$

Similarly, the distances between other nodes in slot n are expressed as

$$d_{JD}[n] = \sqrt{(X_J - X_D[n])^2 + (Y_J - Y_D[n])^2}, \quad (2)$$

$$d_{SU}[n] = \sqrt{(X_S - X_U[n])^2 + (Y_S - Y_U[n])^2 + H^2}, \quad (3)$$

$$d_{UD}[n]=\sqrt{(X_D[n]-X_U[n])^2+(Y_D[n]-Y_U[n])^2+H^2}, \quad (4)$$

$$d_{JU}[n] = \sqrt{(X_J - X_U[n])^2 + (Y_J - Y_U[n])^2 + H^2}. \quad (5)$$

In order to model the movement of the IRS-UAV, the movement direction and movement distance of IRS-UAV in slot n are denoted as $\mu[n]$ and $d[n]$. Supposing the initial position coordinate of IRS-UAV is $\mathbf{W}_U[0] = \{X_0, Y_0\}$. Then the coordinates of IRS-UAV in slot n can be described as

$$X_U[n] = X_0 + \sum_{i=1}^{n} (\cos \mu[i]) d[i], \quad (6)$$

$$Y_U[n] = Y_0 + \sum_{i=1}^{n} (\sin \mu[i]) d[i]. \quad (7)$$

In each time slot, we assume that the maximum velocity of IRS-UAV is $v_{\max}$. Thus, the maximum distance of IRS-UAV in each time slot is $d_{\max} = v_{max}\tau$. Due to the IRS-UAV moves in the specified area, the trajectory of IRS-UAV follows the following constraints.

$$0 \le X_U[n] \le X_{\max}, \quad (8)$$

$$0 \le Y_U[n] \le Y_{\max}, \quad (9)$$

$$0 \le d[n] \le d_{\max}, \quad (10)$$

$$\mu[n] \in [0, 2\pi), \quad (11)$$

where $X_{\max}, Y_{\max}$ represent the abscissa and ordinate of the specified area boundary.

3.2 IRS-assisted Communication Channel Model

The IRS is regularly arranged with K rows and M columns and assume always keeps parallel to the ground during the flight [36]. $E_{k,m}$ denotes the units cell in the k^{th} row and m^{th} column, which can adjust the amplitude and phase shift of the reflected signal by the IRS controller. Thus the programmable reflection coefficient of $E_{k,m}$ denotes as $\Gamma_{k,m} = Ae^{j\phi_{k,m}}$, where $A \in [0, 1]$ denotes the amplitude and $\phi_{k,m} \in [0, 2\pi)$ denotes the phase shift. We assume each unit cell shares the same amplitude A but different phase shifts $\phi_{k,m}$, which ought to be properly designed. The baseband equivalent channels from S to D and J to D are denoted by $h_{SD}^H \in \mathbb{C}^{1\times1}$ and $h_{JD}^H \in \mathbb{C}^{1\times1}$. The reflecting channels of $S-U-D$ and $J-U-D$ denote as $H_S \in \mathbb{C}^{1\times1}$ and $H_J \in \mathbb{C}^{1\times1}$ [37]. Suffering from severe channel attenuation, we only consider the signals reflected once and ignore the signals reflected multiple times [14]. The signal received by the mobile user in slot n can be expressed as

$$\begin{aligned} y = &\sqrt{P_S}(H_S + h_{SD}^H)x_S \\ &+ \sqrt{P_J}(H_J + h_{JD}^H)x_J + z, \end{aligned} \quad (12)$$

where x_S and x_J denote signals sent by base station and jammer, P_S and P_J denote the transmit power of base station and jammer, z denotes i.i.d. additive white Gaussian noise (AWGN) at mobile user with zero mean and variance δ^2.

Thus, the Signal to Interference plus Noise Ratio (SINR) at mobile user can be expressed as

$$\mathrm{SINR} = \frac{P_S \left|H_S + h_{SD}^H\right|^2}{P_J \left|H_J + h_{JD}^H\right|^2 + \delta^2}. \quad (13)$$

As shown in Figure 2, we establish a rectangular coordinate system with the center of the IRS as the origin. Assuming the wavelength of the signal is λ. The sizes of each unit cell are d_x and d_y, which are usually of subwavelength scale within the range of $\frac{\lambda}{10}$ and $\frac{\lambda}{2}$. Since IRS is mounted on UAV, the channels

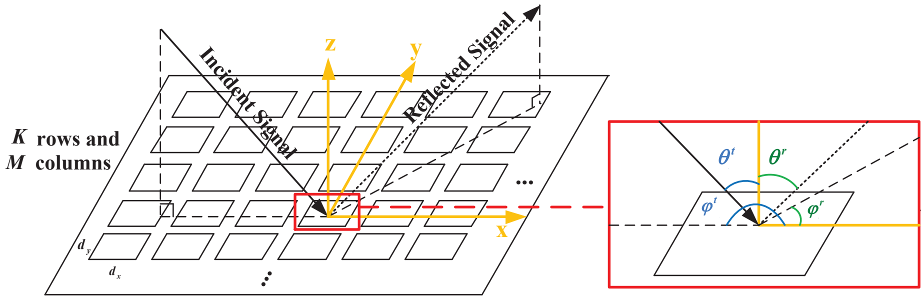

Figure 2. *IRS-assisted wireless communication.*

between U and ground nodes can be modeled as free-space channel models. Differently, we use the far-field free-space channel models in [37], which considers the physics and electromagnetic nature of the RISs. The reflecting channel can be expressed as

$$H = \sqrt{\frac{G_t G_r G M^2 K^2 d_x d_y \lambda^2 F(\theta_t, \varphi_t) F(\theta_r, \varphi_r) A^2}{64\pi^3 d_1^2 d_2^2}} \times Z(\theta_t, \varphi_t, \theta_r, \varphi_r), \tag{14}$$

where G_t, G_r and G denote the antenna gain of transmitter, receiver and unit cell of IRS, d_1 and d_2 is the distance from the center of the IRS to the transmitter and to the receiver. θ_t, φ_t, θ_r and φ_r denote the angles between incident signal and x axis, incident signal and z axis, reflection signal and x axis and reflection signal and z axis, respectively. $F(\theta, \varphi)$ is the normalized power radiation pattern of the unit cell. $Z(\theta_t, \varphi_t, \theta_r, \varphi_r)$ is the beamforming direction error factor, which can expressed as

$$\begin{aligned} &Z(\theta_t, \varphi_t, \theta_r, \varphi_r) \\ &= \frac{\sin c(\frac{M\pi}{\lambda}(\sin\theta_t \cos\varphi_t + \sin\theta_r \cos\varphi_r + \delta_1) d_x)}{\sin c(\frac{\pi}{\lambda}(\sin\theta_t \cos\varphi_t + \sin\theta_r \cos\varphi_r + \delta_1) d_x)} \\ &\times \frac{\sin c(\frac{K\pi}{\lambda}(\sin\theta_t \sin\varphi_t + \sin\theta_r \sin\varphi_r + \delta_2) d_y)}{\sin c(\frac{\pi}{\lambda}(\sin\theta_t \sin\varphi_t + \sin\theta_r \sin\varphi_r + \delta_2) d_y)}, \end{aligned} \tag{15}$$

where $\delta_1(m - \frac{1}{2})d_x + \delta_2(k - \frac{1}{2})d_y = \frac{\lambda\phi_{k,m}}{2\pi}$. According to [37], when the beamforming direction is aligned with the receiver, $\delta_1 = -\sin\theta_t \cos\varphi_t - \sin\theta_r \cos\varphi_r$ and $\delta_2 = -\sin\theta_t \sin\varphi_t - \sin\theta_r \sin\varphi_r$. Therefore, the corresponding design of $\phi_{k,m}$ can be expressed as

$$\begin{aligned} \phi_{k,m} = \mathrm{mod}(&-\frac{2\pi}{\lambda}((\sin\theta_t \cos\varphi_t + \sin\theta_r \cos\varphi_r)(m - \frac{1}{2})d_x \\ &+ (\sin\theta_t \sin\varphi_t + \sin\theta_r \sin\varphi_r)(k - \frac{1}{2})d_y), 2\pi). \end{aligned} \tag{16}$$

The reflection coefficient of IRS is made up of $\phi_{k,m}$ of all the reflection units. Since the phase shift design of each reflecting unit of the large-scale IRS requires a huge amount of calculation, it is difficult to match the real-time change of the channel state. So we consider using the actual physical model, the mapping of the intelligent reflecting surface reflection coefficient and the beamforming direction relationship, reducing the computational complexity of passive beamforming for large-scale IRS. In this scenario, the unknown position of the jammer brings uncertainty to the passive beamforming optimization, which is the key issue we need to solve.

3.3 Problem Formulation

In this paper, we aim to maximize the achievable rate at mobile user by jointly optimizing the trajectory $\mathbf{W}_U[n]$ of UAV and reflection coefficient $\Theta[n] = diag(\phi_{1,1}, ..., \phi_{k,m}, ..., \phi_{K,M})$ of IRS. Now, according to Shannon's formula, the joint problem considered in this paper can be mathematically described as

$$(\mathrm{P1}): \max_{\Theta[\mathrm{n}], \mathrm{W}_\mathrm{n}[\mathrm{n}]} \log_2(1 + \mathrm{SINR}), \tag{17}$$

s.t.

$$0 \le X_U[n] \le X_{\max}, \forall n \in \mathcal{N}, \tag{18}$$

$$0 \leq Y_U[n] \leq Y_{\max}, \forall n \in \mathcal{N}, \tag{19}$$

$$0 \leq d[n] \leq d_{\max}, \forall n \in \mathcal{N}, \tag{20}$$

$$\mu[n] \in [0, 2\pi), \forall n \in \mathcal{N}, \tag{21}$$

$$\phi_{k,m}[n] \in [0, 2\pi), \forall n \in \mathcal{N}. \tag{22}$$

In practice, it has some difficulties finding the optimal trajectory and beamforming of IRS-UAV in (P1). First, the achievable communication rate depends on the IRS-UAV trajectory and beamforming in sophisticated manners. Also, the power and location of jammer is unknown and the trajectory of mobile user is unpredictable, which is challenging to optimization. Furthermore, even these information are perfectly known, the resulting optimization problem will be highly non-convex and difficult to be efficiently solved.

IV. DUELING DOUBLE DEEP Q NETWORKS MULTI-STEP LEARNING ALGORITHM FOR TRAJECTORY AND BEAMFORMING OPTIMIZATION

We propose the dueling double deep Q networks multi-step learning algorithm for IRS-UAV trajectory and beamforming optimization to overcome the above issues. The optimization problem is formulated into MDP, and the state, action, and reward are properly designed to reduce the decision space of algorithm. By leveraging the mathematical framework of deep learning (RL) and deep neural networks (DNN), IRS-UAV can learn the environment and select the proper strategy. After the convergence of DNN, the trajectory and beamforming scheme will finally be obtained.

4.1 IRS-UAV Assisted Anti-jamming Communication as MDP

The first step to utilize proposed DRL-based algorithm is to formulate the optimization problem as a MDP.

The MDP can be expressed by $< \mathcal{S}, \mathcal{A}, \mathcal{R}, \mathcal{P} >$, where $\mathcal{S}$ denotes the state space, $\mathcal{A}$ denotes the action space, $r(s_t, a_t, s_{t+1}) \in \mathcal{R}$ denotes the reward of taking action a_t in state s_t and changing state to s_{t+1}. $P(s_{t+1}|s_t, a_t) \in \mathcal{P}$ denotes the probability of taking action a_t in state s_t and changing state to s_{t+1}. Let us denote the strategy of selecting action in a certain state as $\pi = \mathcal{S} \to \mathcal{A}$, and the history of state-action as $\mathbf{h}_t = (s_t, a_t, s_{t-1}, a_{t-1}, ..., s_{t-\infty}, a_{t-\infty})$. For certain s_{t+1}, $\mathbf{h}_t$, and a_t, an MDP should satisfy the Markov property, which is expressed as

$$P(s_{t+1}|\mathbf{h}_t) = P(s_{t+1}|s_t, a_t). \tag{23}$$

The long-term expected cumulative return value function of taking strategy π in state s_t is denoted as $V^{\pi}(s_t)$. The action-value function of taking action a_t in state s_t is denoted as $Q(s_t, a_t)$. An optimal strategy π^* subjecting to $V^{\pi^*}(s_t) = \max_{a_t \in \mathcal{A}} Q(s_t, a_t)$. The optimization objective of MDP is to find the strategy π^* that maximizes the long-term expected cumulative reward, which can be expressed as

$$\max_{\pi} \mathbb{E}\left(\sum_{i=1}^{\infty} \gamma^{i-1} r_{t+i}\right) = \max_{\pi} V^{\pi}(s_t), \tag{24}$$

where $0 < \gamma < 1$ is the discount factor, which reflects the influence of future rewards on the current decision. Now according to the Bellman equation, $\pi^*(s_t)$ can be expressed as

$$\pi^*(s_t) = \arg\max_{a_t} \left[\mathbb{E}(r_{t+1}) + \gamma \sum_{s_{t+1} \in S} P(s_{t+1}|s_t, a_t) V^*(s_{t+1})\right]. \tag{25}$$

Notably, though the dynamic programming method can solve Eq. (25) by recursion, the agent can not obtain the $\Pr(s_{t+1}|s_t, a_t)$ in practice. Moreover, in this paper, we assume that the jammer location is unknown to the agent, leading to the model-free DRL. Considering these in mind, to optimize problem (P1), we first define the state $\mathcal{S}$, action $\mathcal{A}$ and reward $\mathcal{R}$ as follows.

4.1.1 State $\mathcal{S}$

The state $s \in \mathcal{S}$ is composed of locations of the UAV and mobile user. The camera can obtain this information in IRS-UAV and image process technologies.

4.1.2 Action $\mathcal{A}$

The action $a \in \mathcal{A}$ is made up of moving direction μ, moving distance d, and phase shift $\phi_{k,m}$. Since the algorithm mainly deals with discrete variables, we discrete μ, d as finite values. The state of location of IRS-UAV is also discrete. However, when the numbers of IRS unit cells increase, the action space of $\phi_{k,m}$ is huge.

In order to handle this problem, the actions only select the beamforming directions of IRS, θ_t, φ_t, θ_r and φ_r. The corresponding $\phi_{k,m}$ can be easily obtained from Eq. (16) by determining the beamforming direction. In this paper, the beamforming has two options to enhance the anti-jamming performance:

- Enhance Useful Signal: For the useful signal sent by the base station, the unit cells adjust the phase shift of the reflection link ($S-U-D$) to be in phase with the phase change of the direct link ($S-D$).
- Cancel Jamming Signal: For the jamming signal sent by the jammer, the unit cells adjust the phase change of the reflection link ($J-U-D$) to reverse the phase change of the direct link ($J-D$).

Since the beamforming direction of $S-U-D$, θ_{St}, φ_{St}, θ_{Sr} and φ_{Sr} can be obtained according to the location coordinates, the programmable reflection coefficient of enhancing useful signal can be drawn from Eq. (16).

Similarly, the beamforming direction of $J-U-D$, θ_{Jr} and φ_{Jr} can be obtained. However, θ_{Jt} and φ_{Jt} are unknown to IRS-UAV. Therefore, we discrete the θ_{Jt} and φ_{Jt} as finite action choices for optimization. In the action selection, the IRS-UAV selects the beamforming direction and chooses the corresponding programmable reflection coefficient $\phi_{k,m}$, which significantly reduces the action space when the number of unit cells is enormous. It is worth noting that the above beamforming direction is discretized in the DQN algorithm.

4.1.3 Reward $\mathcal{R}$

the reward $r \in \mathcal{R}$ provides the evaluation of optimized actions at the last moment. Therefore, we choose the achievable rate at the mobile user as the reward, which can be express as

$$r = \log_2\left(1 + \frac{P_S \left|H_S + h_{SD}^H\right|^2}{P_J \left|H_J + h_{JD}^H\right|^2 + \delta^2}\right). \tag{26}$$

The action a_t changing state s_{t+1} is only determined by state s_t, which ignores the influence of past states $\mathbf{h}_t$. Thus, the set of $< \mathcal{S}, \mathcal{A}, \mathcal{R}, \mathcal{P} >$ satisfies Eq. (23), which can be optimized by DRL algorithm.

4.2 3-DQN Multi-step Learning Algorithm for IRS-UAV Assisted Anti-jamming Communication

3-DQN multi-step learning algorithm combines the advantages of deep learning and reinforcement learning (RL),which improves the ability of RL to deal with problems in unknown and dynamic environments. On the other hand, considering the optimization problem of trajectory and beamforming, we utilize the dueling layer and multi-step learning to improve the traditional double deep Q networks (DDQN) algorithm.

As shown in Figure 3, the 3-DQN multi-step learning networks are composed of neural networks and experience replay memory. The neural networks are made up of double dueling networks, which are evaluation net and target net, respectively. The dueling layer is added to separately estimating the state value and state-dependent action advantages. And then smartly combine them via an aggregating layer as the estimate Q-value [38]. In trajectory and beamforming optimization, the values of states differ greatly and the gaps between different actions in the same state are small. Such a process can reduce the estimation error caused by a slight difference of rewards without dueling layers.

Note that the double dueling networks have the same structure as neural networks. The evaluation net updates in real-time, and the target net updates by duplicating the parameters of the evaluation net periodically. The updating objectives of neural networks are to minimize loss function, which can be expressed as

$$L(\delta) = \mathbb{E}\left[r + \gamma Q(s', a'|\delta') - Q(s, a|\delta)\right], \tag{27}$$

where $a' = \arg\max_a Q(s', a|\delta)$ is defined as the best-reward action selected from evaluation net with parameters δ. $Q(s', a'|\delta')$ is the Q-value of target net with δ'. Utilizing different networks for action

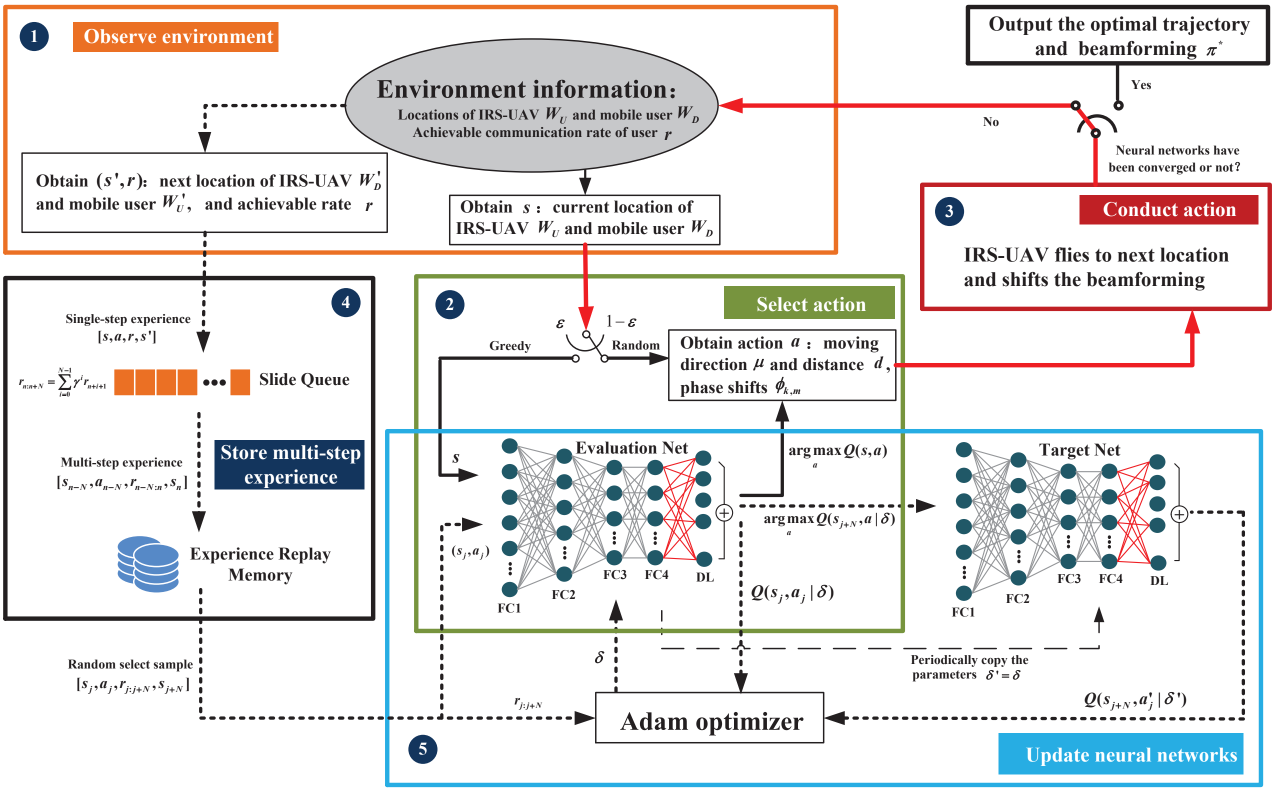

Figure 3. *The process of 3-DQN multi-step learning algorithm.*

selection and Q-value estimation can greatly reduce the overestimation caused by the maximum operation [39]. Moreover, by fixing the parameters of target neural network, the loss function can be kept relatively stable, which significantly improves the speed of convergence.

Notably, we introduce the experience replay memory to store the experience. The training samples of updates can be randomly extracted from the experience replay memory, reducing the correlation between training data and improving the learning ability of neural networks. In trajectory planning, not only the value of single step needs to be considered but also the value of trajectory is essential. Thus, the experience replays memory stores the multi-step experience. The N-step reward is defined as

$$r_{n:n+N} = \sum_{i=0}^{N-1} \gamma^i r_{n+i+1}. \tag{28}$$

To this end, the loss function Eq. (27) is changed as

$$L(\delta) = \mathbb{E}\left[r_{n:n+N} + \gamma^N Q(s', a'|\delta') - Q(s, a|\delta)\right], \tag{29}$$

as discussed in [40], the learning efficiency of algorithm can be significantly enhanced by appropriately choosing N.

The process of dueling double deep Q networks multi-step learning algorithm is shown in Figure 3.

- Observe environment: when the algorithm executes, the state s is obtained from the environment.
- Select action: the action a is chosen according to $\varepsilon - greedy$ strategy, which means that there is a probability of ε to choose the action a with the max Q-value, and the probability of $1-\varepsilon$ to choose a random action. Note that the Q-values are estimated by inputting the obtained state s into evaluation net. As the episode increases, the ε will increase accordingly. This design strikes a balance between exploration and exploitation.
- Conduct action: the IRS-UAV operates selected action a, then obtains the reward r and next state s' from the environment, the cycle between neural

Algorithm 1. *3-DQN multi-step learning algorithm for IRS-UAV enhanced anti-jamming networks.*

1: Initialize the structure and parameters of valuation net and target net;
2: Initialize experience replay memory;
3: **for** *epoch* $= 1, 2, ..., N_{ep}$ **do**
4: $n = 0$;
5: Initialize state s_0;
6: Initialize a window queue W with capacity N_{ns}
7: **while** $n < N$ **do**
8: Obtain state s_n ;
9: Select $a_n = \arg\max_{a_n} Q(s_n, a_n)$with probability ε;
10: Randomly select a_n with probability $1 - \varepsilon$;
11: Execute a_n;
12: Obtain r_n and s_{n+1} ;
13: Store $[s_n, a_n, r_n, s_{n+1}]$ in sliding window queue W;
14: **if** samples of sliding window queue W $> N_{ns}$ **then**
15: Calculate the multi-step reward based on Eq. (28), and store the $[s_{n-N_{ns}}, a_{n-N_{ns}}, r_{(n-N_{ns}):n}, s_{\mathrm{n}}]$ in experience replay memory;
16: **end if**
17: **if** samples of experience replay memory $>$ N_{me} **then**
18: Randomly select $[s_j, a_j, r_{j:j+N_{ns}}, s_{j+N_{ns}}]$ from experience replay memory;
19: Start training evaluation net based on the gradient of loss function Eq. (29);
20: update target net periodically;
21: **end if**
22: $n = n + 1$;
23: Increase ε;
24: **end while**
25: **end for**

network and environment has been completed. At the same time, the algorithm also gains a single-step experience for training.

- Store multi-step experience: the obtained single-step experiences $[s, a, r, s']$ are continuously stored into the window queue W until it is filled. Then according to Eq. (28), multi-step experience $[s_{n-N}, a_{n-N}, r_{n-N:n}, s_n]$ is calculated and stored in experience replay memory.
- Update neural networks: when experience replay memory is full, the training of neural nets starts. First, the random training sample is extracted from experience replay memory and put into neural networks to get the loss function according to Eq. (29). Secondly, the parameters of evaluation net are updated by Adam optimizer, which aims to minimize the loss function. Finally, the parameters of the evaluation network are copied to the target network after a few cycles. By continuously iterating the above process, the neural networks can fit the environment better. Significantly, ε is slowly increased after every episode to make the algorithm gradually converged.

The pseudo-code of the 3-DQN multi-step learning algorithm is provided in algorithm 1. Steps 1-2 are the initialization of the algorithm. For every episode, steps 5-6 are the initialization of the environment. After steps 8-12, the single-step experience is obtained. Steps 13-17 are the process of getting multi-step experience. The neural networks are updated in steps 18-19. When the update is finished, the ε is increased in the end of each step. Therefore the process of each episode is finished.

Table 1. *Main simulation parameters.*

Notation	Discription	Notation	Discription
K	100	$\mathbf{W}_D[0]$	[1000,1000]
N_{epi}	500	$\mathbf{W}_S$	[10,10]
N_{ns}	10	X_{max}	1000m
N_{me}	10000	Y_{max}	1000m
A	0.9	d_{max}	15m
N_μ	4	γ	0.99
N_d	3	ϵ	0.1
N_θ	4	P_S	80dBm
$\mathbf{W}_U[0]$	[800,100]	σ^2	-105dBm
d_x	0.01m	d_y	0.01m
G_t	21dB	G_r	21dB

V. SIMULATION RESULTS

This section provides comprehensive simulations to evaluate the performance of the proposed algorithm, which are conducted in python 3.7 and TensorFlow 2.0.0. The operating frequency $f = 10.5GHz$, $\lambda = c/f = 0.0286m$ following [37] and this assumption meets the requirement of large-IRS far-field free

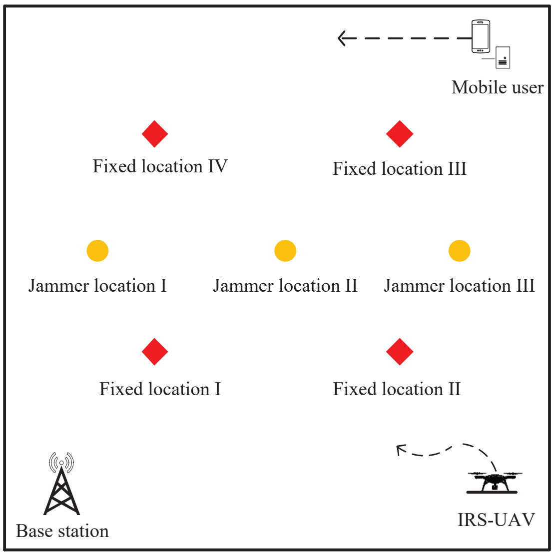

Figure 4. *Top view of locations.*

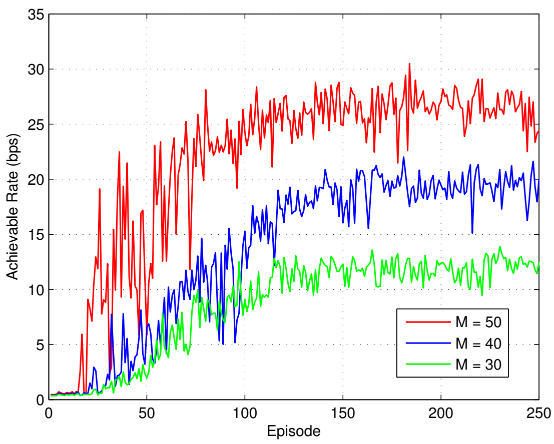

Figure 5. *Achievable rate versus the number of reflecting unit cells on the IRS.*

space model in our scenario. The structures of neural networks are set as [512,256,128,128] fully connected network with dueling layer in the end. The locations of communication parties are shown in Figure 4. The coordinates of jammer location I-III are [200,500], [500,500], [800,500], respectively. We assume the mobile user makes linear uniform motion from $\mathbf{W}_D[0]$, and the IRS-UAV takes off from the $\mathbf{W}_U[0]$. The main simulation parameters are set in table 1.

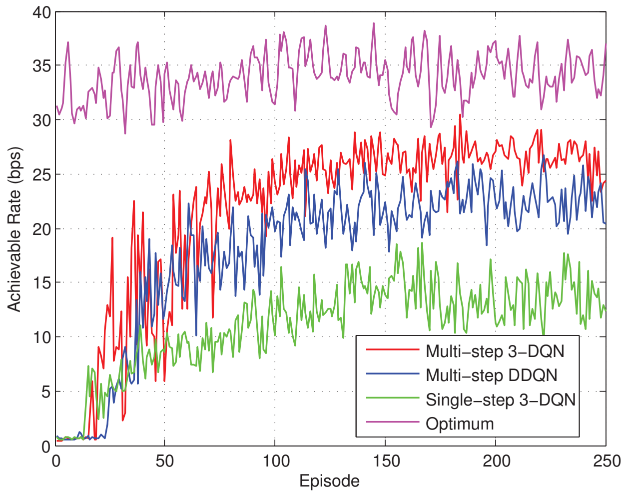

Figure 6. *Achievable rate versus comparison algorithms.*

5.1 Analyses of Algorithm Convergence

Figure 5 shows the achievable rate convergence performance of the proposed algorithm under different numbers of unit cells. The jammer is in jamming location II of Figure 4, and each point on the graph is the average of 1000 episodes. At the beginning of episodes, the agent randomly selects actions from $\mathcal{A}$ and conducts actions to obtain experience. As the increasing of episodes, the probability of randomly choosing actions decreases. The agent, therefore, tends to choose actions with high rewards, which improves the achievable rate. The trajectory and beamforming planning will gradually converge when the random exploration stops. It is shown in Figure 5 that the achievable rate is converged in the end. Additionally, the convergence of the proposed algorithm is independent of the unit number. We can also observe that the achievable rate increases as the number of unit cells on IRS increases, which shows that the anti-jamming performance can be significantly improved by deploying more unit cells.

The 3-DQN algorithm is proposed to satisfy the environment of our scenario. Significantly, the dueling networks and multi-step improvements of the algorithm have better performance in joint optimization of trajectory and passive beamforming. For validating this, we set the comparison algorithms as multi-step DDQN and single-step 3-DQN. The multi-step DDQN and the multi-step 3-DQN are different in the structure of neural networks. The multi-step DDQN is without dueling layers to estimate the slight gaps between actions of the same state. As shown in Figure 6, this drawback of multi-step DDQN is reflected in comparing multi-step 3-DQN and multi-step DDQN. The estimate error of actions reduce the achievable rate at mobile user. We can also tell from the picture that the single-step 3-DQN has even worse performance. Although the single-step 3-DQN has the more accurate Q value, the myopic influence of single-step greatly damages the anti-jamming performance in trajectory optimization. Especially, the IRS-UAV may need to pass through an area of low value in some cases. When the IRS-UAV only considers the value of a single step, it may also abandon the better rewards beyond this area. The optimum curve is the optimal return trajectory and beamforming scheme obtained by using exhaustive search in the recorded experimental data at all times. We can observe that the proposed algorithm can obtain the real-time trajectory and beamforming and has about 80% of the optimum performance.

5.2 Analyses of Algorithm Complexity

According to [41], the complexity of algorithm is

$$Time \sim O(\sum\nolimits_{t_f}(2I_{l_f}E_{lf}-1)), \quad (30)$$

where l_f is the number of fully connected layers, I is the number of inputs, and E is the number of output. The determining factors of complexity and real-time performance are the number of input and output.

In this paper, to reduce the complexity of proposed algorithm and improve the real-time performance, we consider the relationship between beamforming direction and the phase shift of IRS. The action space is significantly reduced by taking the beamforming direction instead of each phase shift of IRS. The number of inputs and outputs before simplification is determined by the number of reflection units, especially in the large-scale IRS. Such calculation amount will affect the real-time performance of passive beamforming.

Take $M = 50$ as an example. The amount of floating-point operations per second required is about 6.7×10^6. After the optimization, the amount of floating-point operations per second required is about 3.7×10^5. The standard multi-core central processing unit can meet the computational load and has excellent real-time performance.

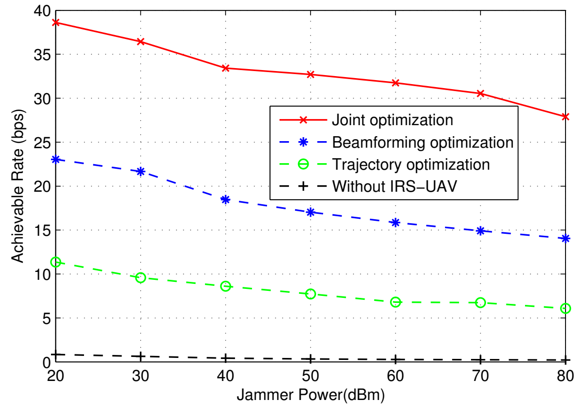

Figure 7. *Achievable rates of different strategies versus the power of jammer.*

5.3 Analyses of Joint Optimization Performance

Figure 7 compares the anti-jamming performance of joint optimization versus jamming power. The column number M is 50, and the jammer is in jamming location II of Figure 4. In order to verify the effectiveness of the joint optimization algorithm. We propose the following heuristic strategies: (I) Proposed algorithm of joint trajectory and passive beamforming optimization, 3-DQN is used to obtain the trajectory and beamforming. (II) Beamforming optimization with the fixed location of IRS-UAV. The fixed location is chosen from Figure 4, with the best performance among the four fixed locations. (III) Trajectory optimization with fixed beamforming. (IV) Communication without the assist of IRS-UAV.

We can first observe that the achievable rates of all strategies decrease as the jamming power increases. The performance gap between the proposed algorithm and beamforming with fixed location generally remains unchanged, validating the advantage of trajectory optimization. Besides, We compare the performance of proposed joint optimization and trajectory optimization only. The performance gap is more extensive than the above comparison. This is because the random beamforming not only damages the anti-jamming performance of IRS but also provides an inaccurate reward which affects trajectory optimization. The three strategies I-III have better performances than strategy IV, which reflects that the application of IRS in anti-jamming has a promising prospect.

In Figure 8, the achievable rates of different strate-

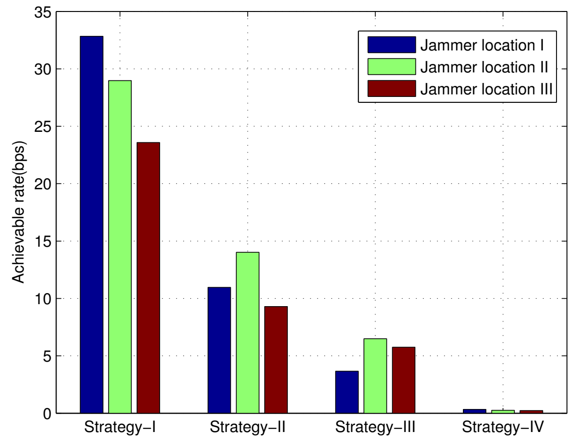

Figure 8. *Achievable rates of different strategies versus the location of jammer.*

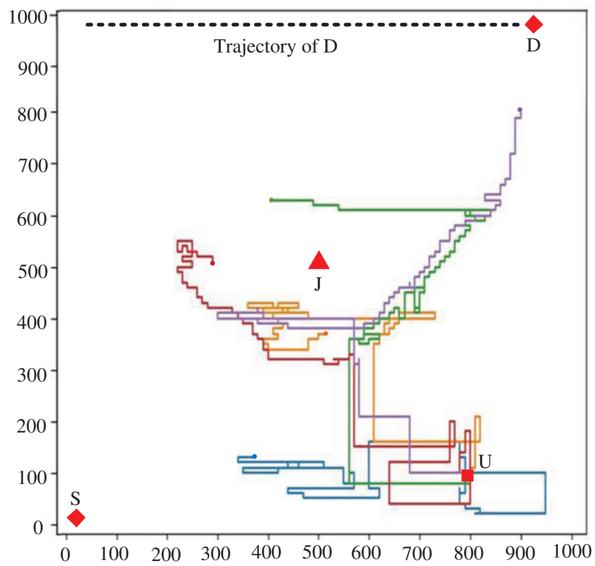

Figure 9. *Trajectory of random exploration (episodes 45-49).*

gies versus the location of the jammer are presented. We can observe that the proposed joint optimization has the best performance compared to other strategies no matter where the jammer is. This indicates that the proposed joint optimization can perform well under imperfect jammer information conditions.

5.4 Analyses of Trajectory Optimization

We draw some trajectories in different episodes to study the rules in trajectory optimization. As shown in Figure 9, the trajectories are chosen from episodes 45-49. These trajectories reflect the random explo-

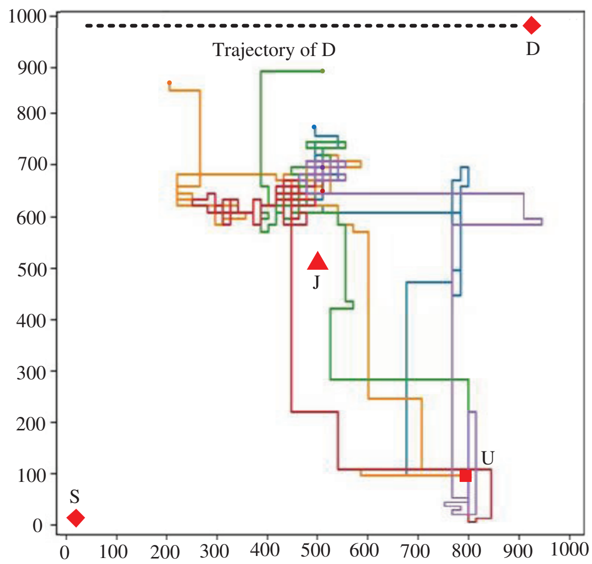

Figure 10. *The trajectory of greedy selection (episodes 95-99).*

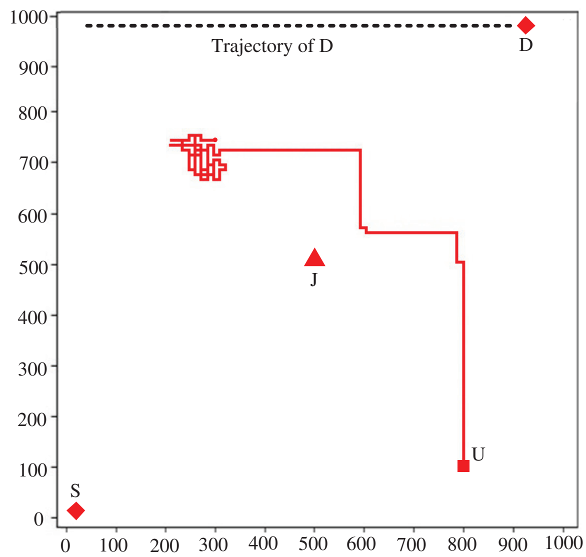

Figure 11. *Optimized trajectory of IRS-UAV (after convergence).*

ration of IRS-UAV at the beginning of the algorithm. At this time, trajectories extend in all directions from the take-off point to collect experience information in various locations. As shown in Figure 10, these trajectories are in episodes 95-99. After enough exploration, the neural networks are trained, and the probability of random exploration is reduced. The IRS-UAV can greedily select the better reward trajectory with proper beamforming. However, the IRS-UAV still has some slight random selection at this time. The optimized trajectory is presented in Figure 11. As we can tell from the trajectory that the IRS-UAV tends to seek the tradeoff of distances between S-D and J-D. Especially, when the jammer is closer to the receiver, the IRS-UAV moves closer to the jammer with a more destructive passive beamforming.

VI. CONCLUSION

Unlike most existing anti-jamming works, we considered the application IRS-UAV to improve the performance against high-power tracking jamming in this paper. A joint trajectory and passive beamforming optimization was investigated to exploit the anti-jamming performance of IRS-UAV fully. Moreover, the joint problem was formulated into MDP and solved by a 3-DQN multi-step learning algorithm to satisfy the imperfect information of jammer and dynamic environment. Simulations were made to verify the effectiveness of the proposed scheme, which showed that the proposed scheme could significantly improve the anti-jamming performance.

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References

[1] Q. Wu and R. Zhang, "Towards smart and reconfigurable environment: Intelligent reflecting surface aided wireless network," *IEEE Communications Magazine*, vol. 58, no. 1, 2019, pp. 106–112.

[2] H. Q. Ngo, E. G. Larsson, *et al.*, "Energy and spectral efficiency of very large multiuser mimo systems," *IEEE Transactions on Communications*, vol. 61, no. 4, 2013, pp. 1436–1449.

[3] S. Cui, A. J. Goldsmith, *et al.*, "Energy-efficiency of mimo and cooperative mimo techniques in sensor networks," *IEEE Journal on selected areas in communications*, vol. 22, no. 6, 2004, pp. 1089–1098.

[4] P. Nayeri, F. Yang, *et al.*, *Reflectarray Antennas: Theory, Designs and Applications*. Wiley Online Library, 2018.

[5] W. Chen, X. Ma, *et al.*, "Sum-rate maximization for intelligent reflecting surface based terahertz communication systems," in *2019 IEEE/CIC International Conference on Communications Workshops in China (ICCC Workshops)*. IEEE, 2019, pp. 153–157.

[6] B. Zheng, Q. Wu, *et al.*, "Intelligent reflecting surface-assisted multiple access with user pairing: Noma or oma?" *IEEE Communications Letters*, vol. 24, no. 4, 2020, pp. 753–757.

[7] S. Y. Park and D. I. Kim, "Intelligent reflecting surface-aided phase-shift backscatter communication," in *2020 14th International Conference on Ubiquitous Information Management and Communication (IMCOM)*. IEEE, 2020, pp. 1–5.

[8] Q. Wu, Y. Zeng, *et al.*, "Joint trajectory and communication design for multi-uav enabled wireless networks," *IEEE Transactions on Wireless Communications*, vol. 17, no. 3, 2018, pp. 2109–2121.

[9] J. Xu, Y. Zeng, *et al.*, "Uav-enabled wireless power transfer: Trajectory design and energy optimization," *IEEE Transactions on Wireless Communications*, vol. 17, no. 8, 2018, pp. 5092–5106.

[10] L. Xie, J. Xu, *et al.*, "Throughput maximization for uav-enabled wireless powered communication networks," *IEEE Internet of Things Journal*, vol. 6, no. 2, 2018, pp. 1690–1703.

[11] X. Zhang and L. Duan, "Fast deployment of uav networks for optimal wireless coverage," *IEEE Transactions on Mobile Computing*, vol. 18, no. 3, 2018, pp. 588–601.

[12] C. Jia, H. Gao, *et al.*, "Machine learning empowered beam management for intelligent reflecting surface assisted mmwave networks," *China Communications*, vol. 17, no. 10, 2020, pp. 100–114.

[13] G. Zhou, C. Pan, *et al.*, "A framework of robust transmission design for irs-aided miso communications with imperfect cascaded channels," *IEEE Transactions on Signal Processing*, vol. 68, 2020, pp. 5092–5106.

[14] J. Dang, Z. Zhang, *et al.*, "Joint beamforming for intelligent reflecting surface aided wireless communication using statistical csi," *China Communications*, vol. 17, no. 8, 2020, pp. 147–157.

[15] S. Fang, Y. Liu, *et al.*, "Wireless communications under broadband reactive jamming attacks," *IEEE Transactions on Dependable and Secure Computing*, vol. 13, no. 3, 2015, pp. 394–408.

[16] Z. Lu, W. Wang, *et al.*, "Modeling, evaluation and detection of jamming attacks in time-critical wireless applications," *IEEE Transactions on Mobile Computing*, vol. 13, no. 8, 2013, pp. 1746–1759.

[17] K. Grover, A. Lim, *et al.*, "Jamming and anti–jamming techniques in wireless networks: a survey," *International Journal of Ad Hoc and Ubiquitous Computing*, vol. 17, no. 4, 2014, pp. 197–215.

[18] Q. Liu, M. Li, *et al.*, "Disrupting mimo communications with optimal jamming signal design," *IEEE Transactions on Wireless Communications*, vol. 14, no. 10, 2015, pp. 5313–5325.

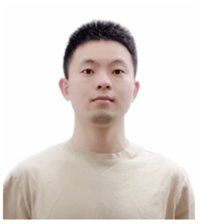
[19] Z. Feng, G. Ren, *et al.*, "Power control in relay-assisted anti-jamming systems: A bayesian three-layer stackelberg game approach," *IEEE Access*, vol. 7, 2019, pp. 14 623–14 636.

[20] D. Yang, G. Xue, *et al.*, "Coping with a smart jammer in wireless networks: A stackelberg game approach," *IEEE Transactions on Wireless Communications*, vol. 12, no. 8, 2013, pp. 4038–4047.

[21] K. Pelechrinis, M. Iliofotou, *et al.*, "Denial of service at-

tacks in wireless networks: The case of jammers," *IEEE Communications surveys & tutorials*, vol. 13, no. 2, 2010, pp. 245–257.
[22] Y. Xu, Y. Xu, *et al.*, "Convert harm into benefit: A coordination-learning based dynamic spectrum anti-jamming approach," *IEEE Transactions on Vehicular Technology*, vol. 69, no. 11, 2020, pp. 13 018–13 032.
[23] N. Qi, W. Wang, *et al.*, "A learning-based spectrum access stackelberg game: Friendly jammer-assisted communication confrontation," *IEEE Transactions on Vehicular Technology*, 2021, pp. 1–1.
[24] Y. Xu, A. Anpalagan, *et al.*, "Decision-theoretic distributed channel selection for opportunistic spectrum access: Strategies, challenges and solutions," *IEEE Communications Surveys & Tutorials*, vol. 15, no. 4, 2013, pp. 1689–1713.
[25] Y. Xu, J. Wang, *et al.*, "Opportunistic spectrum access in cognitive radio networks: Global optimization using local interaction games," *IEEE Journal of Selected Topics in Signal Processing*, vol. 6, no. 2, 2011, pp. 180–194.
[26] Y. Sun, K. An, *et al.*, "Outage constrained robust beamforming optimization for multiuser irs-assisted anti-jamming communications with imperfect csi," 2020.
[27] H. Yang, Z. Xiong, *et al.*, "Intelligent reflecting surface assisted anti-jamming communications: A fast reinforcement learning approach," *arXiv preprint arXiv:2004.12539*, 2020.
[28] B. Liu, Y. Wan, *et al.*, "Robust trajectory and beamforming design for cognitive miso uav networks," *IEEE Wireless Communications Letters*, 2020.
[29] Q. Yuan, Y. Hu, *et al.*, "Joint 3d beamforming and trajectory design for uav-enabled mobile relaying system," *IEEE Access*, vol. 7, 2019, pp. 26 488–26 496.
[30] S. Li, B. Duo, *et al.*, "Reconfigurable intelligent surface assisted uav communication: Joint trajectory design and passive beamforming," *IEEE Wireless Communications Letters*, vol. 9, no. 5, 2020, pp. 716–720.
[31] L. Ge, P. Dong, *et al.*, "Joint beamforming and trajectory optimization for intelligent reflecting surfaces-assisted uav communications," *IEEE Access*, vol. 8, 2020, pp. 78 702–78 712.
[32] X. Liu, Y. Liu, *et al.*, "Machine learning empowered trajectory and passive beamforming design in uav-ris wireless networks," *arXiv preprint arXiv:2010.02749*, 2020.
[33] C. Feng, C. Zhang, *et al.*, "Trajectory and beamforming vector optimization for multi-uav multicast network," in *2019 11th International Conference on Wireless Communications and Signal Processing (WCSP)*. IEEE, 2019, pp. 1–6.
[34] J. Gu, H. Wang, *et al.*, "Energy constrained completion time minimization in uav-enabled internet of things," *IEEE Internet of Things Journal*, 2020.
[35] E. Björnson, Ö. Özdogan, *et al.*, "Intelligent reflecting surface versus decode-and-forward: How large surfaces are needed to beat relaying?" *IEEE Wireless Communications Letters*, vol. 9, no. 2, 2019, pp. 244–248.
[36] Q. Zhang, W. Saad, *et al.*, "Reflections in the sky: Millimeter wave communication with uav-carried intelligent reflectors," in *2019 IEEE Global Communications Conference (GLOBECOM)*. IEEE, 2019, pp. 1–6.
[37] W. Tang, M. Z. Chen, *et al.*, "Wireless communications with reconfigurable intelligent surface: Path loss modeling and experimental measurement," *IEEE Transactions on Wireless Communications*, 2020.
[38] Z. Wang, T. Schaul, *et al.*, "Dueling network architectures for deep reinforcement learning," in *International conference on machine learning*. PMLR, 2016, pp. 1995–2003.
[39] H. Van Hasselt, A. Guez, *et al.*, "Deep reinforcement learning with double q-learning," *arXiv preprint arXiv:1509.06461*, 2015.
[40] R. S. Sutton and A. G. Barto, *Reinforcement learning: An introduction*. MIT press, 2018.
[41] K. He and J. Sun, "Convolutional neural networks at constrained time cost," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2015, pp. 5353–5360.

Biographies



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